Tutorial 12: The residue theorem

Evaluate the following integrals $\oint_C f(z)\,dz$ by identifying the singularities and then using the residue theorem.

1. $f(z) = \dfrac{1}{2e^z + 1}$ and C is the circle $|z| = 4$.
2. $f(z) = \dfrac{1}{e^z - 1}$ and C is the circle $|z - i\pi| = 4$.
3. $f(z) = \dfrac{1}{z - \sin z}$ and C is the circle $|z| = r$ where r is very small.
4. $f(z) = \dfrac{z}{\sin z}$ and C is the circle $|z - 1| = 3$.
5. $f(z) = \dfrac{1}{z^2 \sin z}$ and C is the circle $|z + 1| = 3$.
6. $f(z) = \dfrac{1}{z(1 + \ln(1 + z))}$ and C is the circle $|z| = \frac{3}{4}$.

Tutorial 13: Heaviside's formula

Confirm that the following functions have simple poles and use Heaviside's formula to find the residues to evaluate the following integrals:

1. $\oint \dfrac{1}{z \sin z} dz$ and C is the circle $|z - 3| = 2$.
2. $\oint \dfrac{1}{z\,(e^z - 2)} dz$ and C is the unit circle.
3. $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{x^4 + x^3 + x^2 + x + 1}$. Hint: Use a suitable D-shaped contour